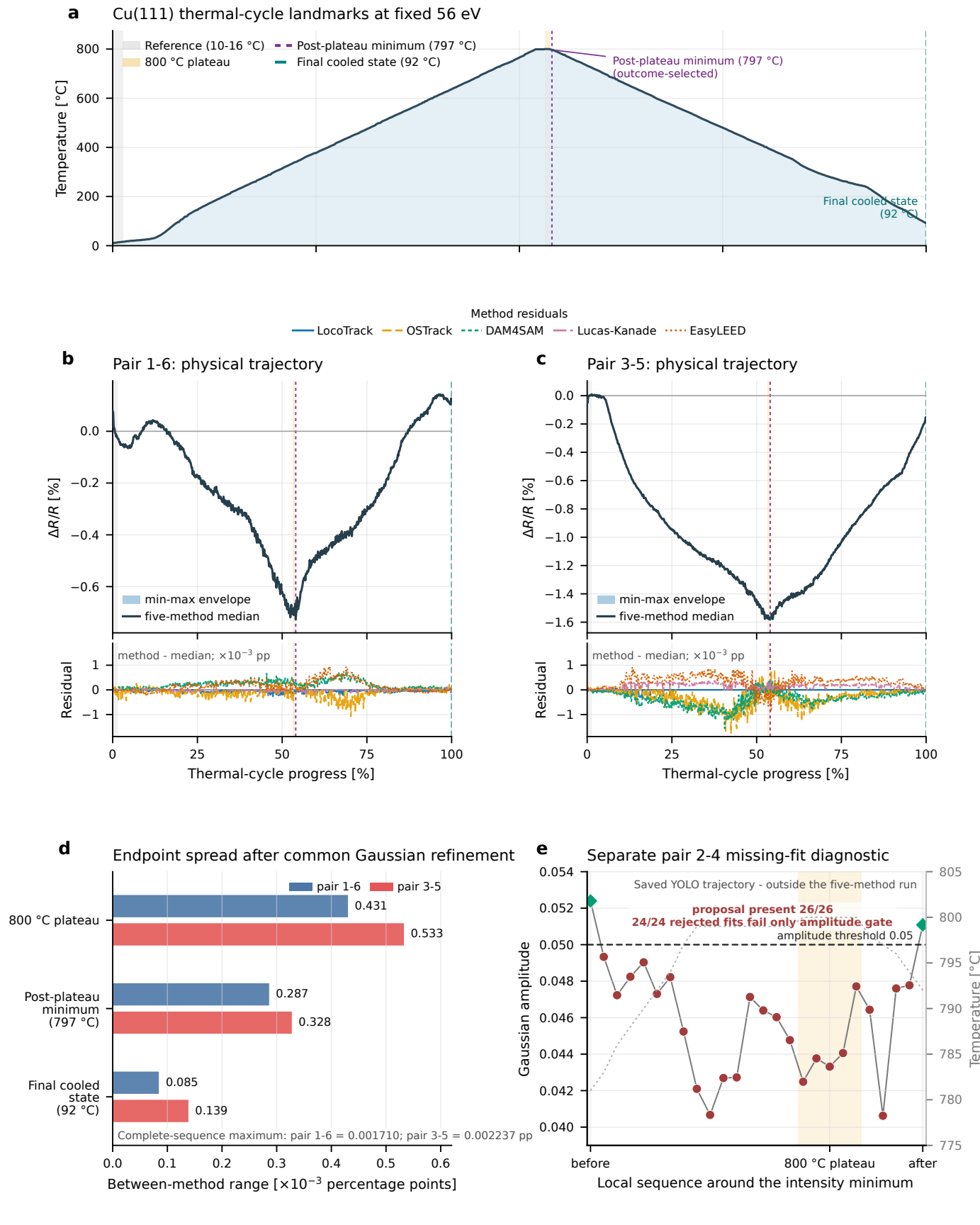

a
Cu(111) thermal-cycle landmarks at fixed 56 eV
Reference (10-16 °C)
800 °C plateau
Post-plateau minimum (797 °C)
Final cooled state (92 °C)
Post-plateau minimum (797 °C)
(outcome-selected)
Final cooled state
(92 °C)
Temperature [°C]
Method residuals
LocoTrack
OSTrack
DAM4SAM
Lucas-Kanade
EasyLEED
b
Pair 1-6: physical trajectory
ΔR/R [%]
min-max envelope
five-method median
method - median; ×10⁻³ pp
Residual
Thermal-cycle progress [%]
c
Pair 3-5: physical trajectory
ΔR/R [%]
min-max envelope
five-method median
method - median; ×10⁻³ pp
Residual
Thermal-cycle progress [%]
d
Endpoint spread after common Gaussian refinement
pair 1-6
pair 3-5
800 °C plateau
0.431
0.533
Post-plateau minimum (797 °C)
0.287
0.328
Final cooled state (92 °C)
0.085
0.139
Complete-sequence maximum: pair 1-6 = 0.001710; pair 3-5 = 0.002237 pp
Between-method range [×10⁻³ percentage points]
e
Separate pair 2-4 missing-fit diagnostic
Saved YOLO trajectory - outside the five-method run
proposal present 26/26
24/24 rejected fits fail only amplitude gate
amplitude threshold 0.05
Gaussian amplitude
Temperature [°C]
before
800 °C plateau
after
Local sequence around the intensity minimum